HTML TUTORIALS
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WEB DESIGNING

What is Web Designing?

Web designing is about making a website look good and easy to use. It includes layout design, color selection, font styles, images, and user-friendly structure.

- Web design focuses on **look and user experience**
- Web development focuses on **coding and functionality**
- A web designer uses tools like **HTML, CSS, and sometimes JavaScript**
- The goal of web design is to make a website **attractive, easy to use, and responsive**

Which technology use in web designing?

- HTML
- CSS
- Bootstrap
- JavaScript
- jQuery

What is HTML?

HTML is the standard markup language used to create web pages and display them in a web browser. HTML describes the structure of a web page using elements.

HTML stands for **Hyper Text Markup Language**. It is called a markup language because it uses tags to mark and structure content such as headings, paragraphs, links, images, and more.

- HTML stands for **Hyper Text Markup Language**
- Html like a skeleton of a website.
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as
  - This is a heading
  - This is a paragraph
  - This is a link
- Without html you cannot create a website
- HTML is based on **elements and tag**

Sample Example of HTML:

```
<!DOCTYPE html>
<html>
<head>
   <title>My First Web Page</title>
</head>
<body>
   <h1>This is a Heading</h1>
   <p>This is a Paragraph.</p>
</body>
</html>
```

Explanation:

- <html> is the root element
- <head> contains page information
- <title> set the page title
- <body> contains visible content
- <h1> is a heading
- <p> is a paragraph

What is Tag?

In html we only use tags for create a website or design a website. It is use in < and > sign. Like <tag>

If any tag open so this is compulsory it will be close.

What is Element?

An HTML element is the basic building block of an HTML document. It is used to define the structure and content of a web page.

An HTML element usually consists of:

- An opening tag
- Content
- A closing tag

HTML elements tell the browser how to display the content on the web page.

<tagname> Content goes here </tagname>

Example:
<p>This is a paragraph</p>

In this example:

- <p> is the opening tag
- This is a paragraph is the content
- </p> is the closing tag
- Together, they form an **HTML element**

What is Text Editor?

- Learn HTML Using Notepad or any Text Editor
- Web pages can be created and modified by using professional HTML editors.
- However, for learning HTML we recommend a simple text editor like Notepad (PC).
- We believe in that using a simple text editor is a good way to learn HTML.

Create One Html Document using Notepad.

Define Text Editor?

Text editor is the software for create a document.

Some Text Editors:

- Notepad
- Sublime
- Dreamweaver
- notepad++
- Atom
- **VS (visual studio) code**

Text Editor are also called IDE (**Integrated Development Environment**).

Two Types of HTML Tags:

In HTML, there are two main types of tags:

1. Paired Tags (Container Tags)

2. Unpaired Tags (Empty Tags)

1. Paired Tags (Container Tags):

Paired tags are HTML tags that have both an **opening tag** and a **closing tag**. The content is written between these tags.

Example:

<h1>This is a Heading</h1>
<p>This is a Paragraph</p>
This is a Link

Explanation:

- <h1> is the opening tag and </h1> is the closing tag
- <p> is the opening tag and </p> is the closing tag
- The content is written between opening and closing tags
- These tags are also called **container tags** because they contain content

2. Unpaired Tags (Empty Tags):

Unpaired tags are HTML tags that **do not have a closing tag**. These tags do not contain content and are self-closing.

Example:

<hr>
<img src="image.jpg" alt="Image">
<input type="text">

Explanation:

-
 is used for line break
- <hr> is used for horizontal line
- <img> is used to display an image
- <input> is used to create input fields
- These tags do not need closing tags

WHAT IS HTML ATTRIBUTES?

HTML attributes provide additional information about HTML elements. Attributes are used to add extra details to an element, such as links, styles, IDs, classes, and more.

- All HTML elements can have **attributes**
- Attributes provide **additional information** about elements
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**
- Attributes make HTML elements more useful and flexible

Example: Visit Website

Common Attributes:

Attribute	Description
href	Specifies the URL of a link
src	Specifies the path of an image
alt	Provides alternative text for an image
title	Shows extra information when mouse hovers
style	Adds inline CSS style
id	Gives a unique ID to an element
class	Groups elements for CSS and JavaScript

What is Comment?

An HTML comment is a piece of code that is **not displayed in the browser**. Comments are used to write notes, explanations, or reminders inside the HTML code. They help developers understand and document the code.

Comments are ignored by the browser, which means they do not affect the webpage layout or design.

Syntax:

```
<!--  Write your comments here -->
```

Example of HTML Comments:

```
<!-- This is heading section -->
<h1>Welcome to My Website</h1>
```

Important Points About HTML Comments:

- Comments are not displayed in the browser

- Comments are not displayed in the browser
- Comments are used to explain the code
- Comments help in code documentation
- Comments are ignored by the browser
- Comments start with <!-- and end with -->

Entity Names

HTML entities are used to display reserved characters and special symbols in a web page. Some characters are reserved in HTML (like < and >), so we use entity names to display them correctly.

Entities start with & and end with ; .

Common HTML Entities:

Entity Name	Description	Symbol
** **	Non-breaking space	(space)
<	Less than	<
>	Greater than	>
&	Ampersand	&
"	Double quotation mark	"
'	Single quotation mark	'
¢	Cent sign	¢
£	Pound sign	£
¥	Yen sign	¥
€	Euro sign	€
©	Copyright	©
®	Registered trademark	®

Example:

<p>This is less than sign: <</p>
<p>This is greater than sign: ></p>
<p>This is ampersand sign: &</p>
<p>This is copyright: ©</p>
<p>This is euro sign: €</p>

What are Semantic Elements?

A semantic element is an HTML element that clearly describes its meaning to both the browser and the developer. In other words, it tells what type of content it contains and what its purpose is.

For Example: For example, instead of just being a generic container, a semantic element clearly defines its role in the structure of a webpage.

Example of Semantic Elements:

All elements are Semantic elements that clearly define their content and purpose:

Example of Non Semantic Elements:

There only 2 Non-semantic elements that do not describe their content clearly:

- <div> it is block element
- <span> it is inline element

These elements do not provide any information about what they contain.

Difference of Semantic & Non Semantic Elements:

Semantic Elements	Non Semantic Elements
They have meaning	They have no meaning
They clearly describe the content inside them	Then can be used for any type of content
They help structure the webpage in a meaningful way	They rely on classes or IDs for styling and structure
They improve accessibility & SEO	They do not describe their purpose

HTML Layout Elements:

Html Lists:

HTML lists are used to group related items together in a structured way. HTML provides three types of lists:

1. Unordered List
2. Ordered List
3. Description List

1. Unordered HTML List

The <ul> element represents an **unordered list** of items, where the order of the items is not important. The list items are usually displayed with bullet points you can change bullets using CSS.

 – List Item:

The <li> element represents an item in a list. It is used inside <ul> or <ol>.

Example:

```
<ul>
  <li>Apple</li>
  <li>Banana</li>
  <li>Mango</li>
</ul>
```

2. Ordered HTML List

The <ol> element represents an ordered list of items, where the order of the items is important. The list items are usually displayed with numbers you can change bullets using CSS. Ordered list is also use for recipe steps where order matter.

Example:

```
<ol>
  <li>Wake up</li>
  <li>Brush teeth</li>
  <li>Go to school</li>
</ol>
```

Nested List in HTML:

A nested list is a list inside another list. It is used to create sub-items under a main list item.

A nested list can be created by placing a <ul> or <ol> inside an <li> element.

Example:

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Banana</li>
      <li>Mango</li>
    </ul>
  </li>

  <li>Vegetables
    <ul>
      <li>Carrot</li>
      <li>Potato</li>
      <li>Onion</li>
    </ul>
  </li>
</ul>
```

3. HTML Description Lists

The <dl> element represents a **description list**, which is used to display terms and their descriptions.

<dt> – Description Term

The <dt> element represents a term or name in a description list.

<dd> – Description Details

The <dd> element represents the description or definition of the term.

Example:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

HTML <a> Tag (Anchor Element):

The <a> (anchor) tag defines a **hyperlink**, which is used to link one page to another, or to different sections within the same page.

The most important attribute of the element is the href attribute, which indicates the link's destination.

Types of Hyperlinks:

External Link (Absolute URL)

```
<a href="about.html">About Us</a>

<a href="https://youtube.com/@theProvidersOfficial">Watch Tutorials</a>

<a href="https://instagram.com/mfarhan.developer">Visit Social Media</a>
```

Target Attribute:

The target attribute specifies where to open the linked document.

1. _self (default) Opens the link in same tab
2. _blank Opens the link in a new tab

Example:

```
<a href="https://youtube.com/@theProvidersOfficial" target="_self"> Open Channel in this tab </a>

<a href="https://youtube.com/@theProvidersOfficial" target="_blank"> Open Channel in new tab </a>
```

Internal Link (Same Page using id)

```
<a href="#about">Go to About Section</a>

<h2 id="about">About Section</h2>
<p>This is the about section of the page.</p>
```

HTML Navigation Menu using <nav>

The `<nav>` element represents a section of a page whose purpose is to provide **navigation links** (menus, table of contents, etc.).

Example:

```
<nav>
<ul>
<li><a href="index.html">Home</a></li>
<li><a href="about.html">About</a></li>
<li><a href="services.html">Services</a></li>
<li><a href="contact.html">Contact</a></li>
</ul>
</nav>
```

HTML Semantic Tags:

<i> Tag – Idiomatic Text

The <i> tag represents text in an alternate voice or mood, such as technical terms, foreign words, thoughts, or idiomatic phrases. The text is usually displayed in italic.

Example:

<p>I looked at it and thought <i>This can't be real!</i></p>

<p>The term <i>bandwidth</i> describes the measure </p>

 Tag – Stylistically Offset Text

The tag is used to draw attention to text without adding extra importance. It is used for keywords, product names, or important words in a paragraph.

Example:

<p> school are <b class="term">chemistry (the study of chemicals and the composition of

substances) and <b class="term">physics (the study of the nature and

properties of matter and energy). </p>

<u> Tag – Unarticulated Annotation

The <u> tag represents text with an unarticulated annotation, usually displayed as underline. It is often used for spelling mistakes or proper names in some cases.

Example:

<p> You could use this element to highlight <u>speling</u> mistakes, so the writer can <u>corect</u> them. </p>

<s> Tag – No Longer Correct Text:

The <s> element to represent things that are no longer relevant or no longer accurate.

Example:

<p><s>There will be a few tickets available at the box office tonight.</s></p>

<p>SOLD OUT!</p>

 Tag – Strong Importance

The tag indicates that the text is very important, serious, or urgent. Browsers display it as bold, but it also adds semantic importance.

Example:

<p> no matter how much he cries, no matter how much he begs: never feed him after midnight. </p>

 Tag – Emphasized Text

The tag is used to emphasize text. It adds stress emphasis, which changes the meaning of the sentence.

Example:

<p>Get out of bed now!</p>

<p>We had to do something about it.</p>

<p>This is not a drill!</p>

<small> Tag – Side Comments / Small Print

The <small> tag represents side comments, copyright text, or small print.

Example:

<p> <small> The content is licensed under a Creative Commons Attribution-ShareAlike 2.5 Generic License.</small> </p>

<del> Tag – Deleted Text:

The <del> tag represents text that has been removed from the document. This can be used when rendering "track changes" or source code diff information

Example:

There is ~~nothing~~ <ins>no code</ins> either good or bad, but

~~thinking~~ <ins>running it</ins> makes it so.

<ins> Tag – Inserted Text:

The <ins> HTML element represents a range of text that has been added to a document. You can use the <del> element to similarly represent a range of text that has been deleted from the document.

Example:

<p>
Old price: ~~$100~~
New price: <ins>$80</ins>
</p>

<sub> Tag – Subscript Text:

The <sub> tag displays subscript text (below the line), often used in chemical formulas.

Example:

```
<p> H<sub>2</sub>O </p>
```

<sup> Tag – Superscript Text

The <sup> tag displays superscript text (above the line), often used in math.

Example:

```
<p> x<sup>2</sup></p>
```

<mark> Tag – Highlighted Text:

The <mark> HTML element represents text which is marked or highlighted for reference or notation purposes due to the marked passage's relevance in the enclosing context.

Example:

```
<p>Search results for "salamander":</p>

<hr />

<p>Several species of <mark>salamander</mark> inhabit the temperate rainforest of  the Pacific Northwest. </p>
```

<header> Tag:

The <header> element represents introductory content for a page or a section. It may include headings, logos, or navigation links.

Example:

```
<header>
<h1>My Website</h1>
<p>Welcome to my homepage</p>
</header>
```

<footer> Tag:

The <footer> element represents the footer for a page or section. It usually contains copyright, links, or contact info.

Example:

```
<footer>
<p>&copy; 2026 The Providers</p>
</footer>
```

<section> Tag:

The <section> element represents a thematic grouping of content, typically with a heading.

Example:

```
<section>
<h2>About Us</h2>
<p>This section contains information about the company.</p>
</section>
```

<aside> Tag:

The <aside> element represents content indirectly related to the main content (like sidebars or ads).

Example:

```
<aside>
<h2>Related Links</h2>
<p>Check out our latest articles.</p>
</aside>
```

<blockquote> Tag:

The <blockquote> element is used for long quotations from another source.

Example:

```
<blockquote>
HTML is the standard markup language for creating web pages.
</blockquote>
<cite>MDN Reference</cite>
```

<q> Tag:

The <q> element is used for short inline quotations.

Example:

```
<p>He said, <q>Learning HTML is easy.</q></p>
```

<cite> Tag:

The <cite> element is used to reference the title of a creative work.

Example:

```
  <blockquote cite="https://www.huxley.net/bnw/four.html">
    <p>
      Words can be like X-rays, if you use them properly—they'll go through
      anything. You read and you're pierced.
    </p>
  </blockquote>
  <p>—Aldous Huxley, <cite>Brave New World</cite></p>
```

<abbr> Tag:

The <abbr> element represents an abbreviation and optionally provides full form using title.

Example:

```
<p><abbr title="HyperText Markup Language">HTML</abbr> is used to build web
pages.</p>
```

<progress> Tag:

The <progress> element represents the progress of a task.

Example:

```
<label>Downloading:</label>
<progress value="70" max="100"></progress>
```

<details> & <summary> Tag:

The <details> element creates a collapsible content section, and <summary> defines its visible heading.

Example:

```
<details>
<summary>What is HTML?</summary>
<p>HTML is used to structure web pages.</p>
<p>HTML is used to structure web pages.</p>
</details>
```

<pre> Tag:

The <pre> element represents preformatted text. It preserves spaces, tabs, and line breaks exactly as written in the HTML code.

Example:

```
<pre>
Hello World
    This text keeps indentation
        And line breaks are preserved
</pre>
```

<code> Tag:

The <code> element represents a fragment of computer code. It is usually displayed in a monospace font.

Example:

```
<p>Use the following HTML tag: <code>&lt;h1&gt;</code></p>
```

Pre & Code Example:

```
<pre>
<code>

&ltscript&gt;

function greet() {
console.log("Hello World");
}

&ltscript/&gt;


</code>
</pre>
```

<article> Tag:

The <article> element represents a self-contained, independent piece of content that can be reused or distributed on its own.

It is commonly used for:

- Blog posts
- News articles
- Forum posts
- Product cards
- Comments

Each `<article>` should make sense even if it is removed from the page.

Example:

```
<article>
<h2>What is HTML?</h2>
<p>HTML is the standard markup language used to create web pages.</p>
</article>
```

Blog Page Example:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Blog Page</title>
</head>

<body>

  <h1>My Blog</h1>
```

```
  <article>
    <h2>Learning HTML</h2>
    <p>HTML helps structure web content using elements and tags.</p>
  </article>

  <article>
    <h2>Learning CSS</h2>
    <p>CSS is used to style and design web pages.</p>
  </article>

  <article>
    <h2>Learning JavaScript</h2>
    <p>JavaScript adds interactivity to web pages.</p>
  </article>

</body>

</html>
```

Basic Semantic Page Layout:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Semantic Layout Example</title>
</head>

<body>

  <header>
    <h1>My Website</h1>
    <p>Learning HTML Semantic Elements</p>
  </header>

  <nav>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">About</a></li>
      <li><a href="#">Services</a></li>
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>

  <main>

    <section>
      <h2>Welcome Section</h2>
      <p>This is the main introduction of the website.</p>
```

```
  </section>

  <article>
   <h2>Latest Blog Post</h2>
   <p>This article explains how semantic HTML improves structure and accessibility.</p>
  </article>

  <aside>
   <h2>Related Info</h2>
   <p>This sidebar contains extra links or advertisements.</p>
  </aside>

 </main>

 <footer>
  <p>&copy; 2026 My Website. All rights reserved.</p>
 </footer>

</body>

</html>
```

Simple Website Layout:

```
<!DOCTYPE html>
<html lang="en">

<head>
 <meta charset="UTF-8">
 <title>Layout Example</title>
</head>

<body>

 <header>
  <h1>My Portfolio</h1>
 </header>

 <nav>
<ul>
  <li> <a href="#">Home</a> </li>
  <li> <a href="#">Projects</a> </li>
  <li><a href="#">Contact</a> </li>
</ul>
</nav>

 <main>
```

```html
    <section>
      <h2>About Me</h2>
      <p>I am a web developer learning semantic HTML.</p>
    </section>

    <section>
      <h2>Projects</h2>
      <p>Here are some of my web development projects.</p>
    </section>

  </main>

  <aside>
    <h2>Skills</h2>
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>
  </aside>

  <footer>
    <p>Contact: example@email.com</p>
  </footer>

</body>

</html>
```

HTML Block and Inline Elements

Every HTML element has a default display value, depending on the type of element. There are two main display types in HTML:

- Block-level elements
- Inline elements

These display types control how elements appear on a web page.

Block-level Elements:

A block-level element always **starts on a new line** and takes up the **full width available** (from left to right). Block-level elements usually have **top and bottom margin**.

Important Points:

- Starts on a new line
- Takes full width
- Has top and bottom margin
- Can contain block and inline elements

Example:

```
<div>Hello World</div>
<p>This is a paragraph</p>
<h1>This is a heading</h1>
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

Common Block Elements:

<div>, <p>, <h1>–<h6>, <ul>, <ol>, <li>, <table>, <section>, <article>, <header>, <footer>, <nav>, <form>, <main>, <pre>, <hr>, <video>

Inline Elements:

An inline element **does not start on a new line** and only takes up **as much width as necessary**. Inline elements are usually used inside block elements.

Important Points:

- Does not start on a new line
- Takes only necessary width
- Does not have top and bottom margin
- Usually used inside block elements

Example:

```
<p>This is a <span>span element</span> inside a paragraph.</p>
```

Common Inline Elements:

`<span>`, `<a>`, `<b>`, `<i>`, `<strong>`, `<em>`, `<img>`, `<input>`, `<label>`, `<small>`, `<sub>`, `<sup>`, `<br>`, `<button>`, `<textarea>`, `<select>`

Html Images:

The <img> element is used to embed an image in a web page.

Important point:

Images are not directly inserted into the page — they are **linked using a URL (source path)**.

The <img> tag is empty, it contains attributes only, and does not have a closing tag.

The <img> tag has two required attributes:

- src - Specifies the path to the image
- alt - Specifies an alternate text for the image

Example:

```
<img src="https://example.com/image.jpg" alt="A beautiful scenery">
```

Example Local Image:

```
<img src="flower_image.jpg" alt="Flower Image">
```

Note: image should be in same directory (folder)

You can control image size using width and height attribute:

```
<img src="flower_image.jpg" alt="Flower Image" width="500" height="600">
```

Html <figure> and <figcaption> Tag:

The <figure> element is used to represent self-contained content, such as images, diagrams, illustrations, or code blocks.

It is often used with <figcaption>, which provides a caption or description for the content inside <figure>.

Why <figure> is used?

Instead of just placing an image alone, <figure> helps to:

- Group image + caption together
- Make content more semantic
- Improve accessibility and structure

Example:

```
<figure>
<img src="nature.jpg" alt="Beautiful mountains" width="400">
<figcaption>Beautiful mountain view during sunrise</figcaption>
</figure
```

<video> Tag:

The <video> element is used to embed video content in a web page.

```
<video src="video.mp4" controls width=”200px” height=”200px” muted autoplay></video>
```

<video> with <source>:

```
<video controls width="400" poster="thumbnail.jpg">
<source src="video.mp4" type="video/mp4">
<source src="video.webm" type="video/webm">
</video>
```

Common Video Attributes:

- controls → shows play/pause buttons
- autoplay → plays automatically
- loop → repeats video
- muted → mutes audio
- poster → shows image before video plays

<audio> Tag:

The <audio> element is used to embed sound content in a web page.

```
<audio src="audio.mp3" controls ></video>
```

<audio> with <source>:

```
<audio controls>
<source src="audio.mp3" type="audio/mpeg">
<source src="audio.ogg" type="audio/ogg">
</audio>
```

Common Audio Attributes:

- controls → shows play/pause buttons
- autoplay → plays automatically
- loop → repeats audio
- muted → mutes audio

Note: autoplay not working in mostly browser if you want this feature you can use javaScript

HTML IFRAMES:

The <iframe> element is used to **embed another HTML page inside the current page**.

It creates a "window" inside your webpage where you can load:

- Websites
- YouTube videos
- Google Maps
- Other web content

Example (Embed a Website):

```
<iframe  src=" http://theproviders.tech/"  >
</iframe>
```

Iframe - Set Height and Width

Use the height and width attributes to specify the size of the iframe.

```
<iframe  src=" http://theproviders.tech/"  width="600"  height="400"> </iframe>
```

Example (Youtube Video Embed):

YouTube provides embed links using `iframe`.

```
<iframe width="560" height="315"
src="https://www.youtube.com/embed/t9KQ4L7l9UI?si=cuFwhACAPg9CYSCu" title="YouTube video
player" frameborder="0" allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope;
picture-in-picture; web-share" referrerpolicy="strict-origin-when-cross-origin" allowfullscreen></iframe>
```

Note: You should run website on any live server because iframe embed not working on local files

Another Example with Anchor:

```
<iframe src="" name="iframe_a" title="Iframe Example"></iframe>

<a href="http://theproviders.tech/" target="iframe_a">The Providers</a>
```

Google Maps Embed:

Go to google map and search any local and click share button and then click embed a map and then copy HTML and paste in your webpage.

HTML TABLES:

The <table> element is used to display tabular data, meaning data arranged in rows and columns.

A table is made using multiple semantic sub-elements:

- <table> → main container
- <tr> → table row
- <th> → table header cell
- <td> → table data cell
- <thead>, <tbody>, <tfoot> → structure sections (optional but recommended)

Simple Table Example:

```
<table border="1">

<tr>
<th>Name</th>
<th>Age</th>
<th>City</th>
</tr>

<tr>
<td>Ali</td>
<td>20</td>
<td>Karachi</td>
</tr>

<tr>
<td>Ahmed</td>
<td>22</td>
<td>Lahore</td>
</tr>

</table>
```

What is Table Cell?

Each table cell is defined by a <td> and a </td> tag.

Everything between <td> and </td> are the content of the table cell.

Table data elements are the data containers of the table.
They can contain all sorts of HTML elements; text, images, lists, other tables, etc.

What is Table Row?

tr stands for table row.

You can have as many rows as you like in a table, just make sure that the number of cells are the same in each row.

Table Header

Sometimes you want your cells to be headers, in those cases use the <th> tag instead of the <td> tag.

By default, the text in <th> elements are bold and centered, but you can change that with CSS.

Table Attributes:

border attribute use in table tag

Colspan & Rowspan Attribute

Colspan: Add two or more columns cell in one cell

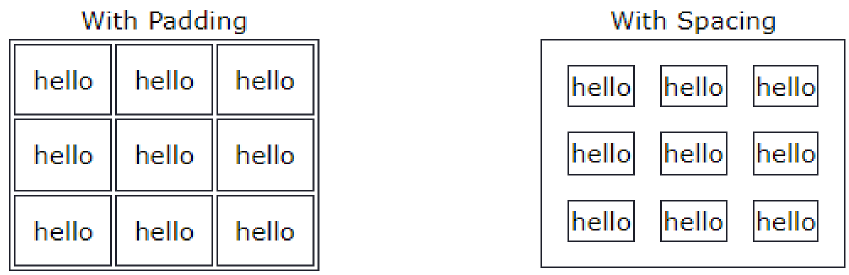
Rowspan: two or more rows cell in one row

Example:

```
<table border=”1”>
  <tr>
    <th colspan="2">Name</th>
    <th>Age</th>
  </tr>
<tr>
    <td rowspan="3">Jill</td>

    <td>Smith</td>
    <td>50</td>
  </tr>
</table>
```

Padding and Spacing Attributes in Table:

HTML tables can adjust the padding inside the cells, and also the space between the cells.



Example:

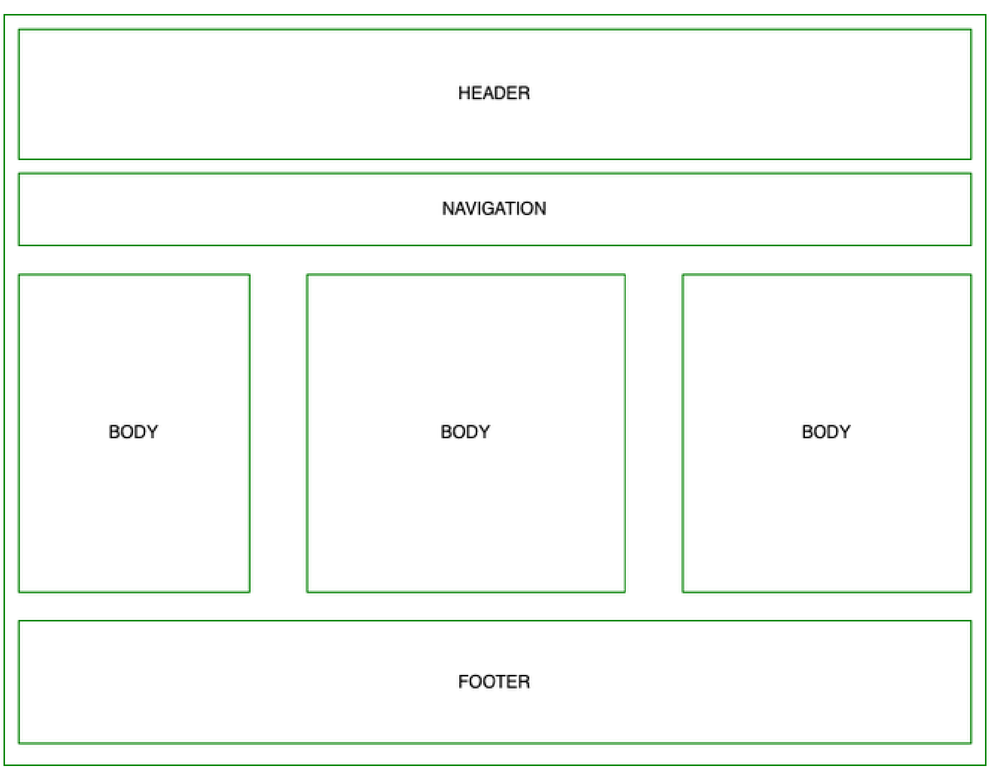
```
<table border="1" cellpadding="10px" cellspacing="0px" width="100%">
    <tr height="80px" align="center" bgcolor="red">
      <td colspan="2">Header</td>
    </tr>
    <tr height="60px" align="center" bgcolor="yellow">
      <td colspan="2">Navigation</td>
    </tr>
    <tr height="430px" align="center" bgcolor="green">
        <td width="75%">Content</td>
        <td>Sidebar</td>
    </tr>
    <tr height="60px" align="center" bgcolor="blue">
      <td colspan="2">Footer</td>
    </tr>
  </table>
```

Complete Table Example:

```
<table border="1">

<thead>
<tr>
<th>Name</th>
<th>Subject</th>
<th>Marks</th>
</tr>
</thead>

<tbody>
```

```
<tr>
<td>Ali</td>
<td>HTML</td>
<td>85</td>
</tr>

<tr>
<td>Sara</td>
<td>CSS</td>
<td>92</td>
</tr>

<tr>
<td>Ahmed</td>
<td>JavaScript</td>
<td>88</td>
</tr>
</tbody>

<tfoot>
<tr>
<td colspan="2">Total Average</td>
<td>88.3</td>
</tr>
</tfoot>

</table>
```

HTML TABLE BASIC LAYOUT (Practice):



HTML FORMS:

HTML Forms are required, when you want to collect some data from the site visitor. For example, during user registration you would like to collect information such as name, email address, credit card, etc.

The <form> Element:

The HTML <form> element is used to create an HTML form. The <form> element is a container for different types of input elements, such as: text fields, checkboxes, radio buttons, submit buttons, etc.

The Html Form Elements:

1. **<input>**
2. **<label>**
3. **<datalist> use with input attribute list.**
4. **<select>**
5. **<option>**
6. **<optgroup>**
7. **<textarea>**
8. **<button>**
9. **<fieldset>**
10. **<legend>**

Autocomplete Attribute:

The autocomplete attribute specifies whether a form should have autocomplete on or off.

By default, autocomplete attribute value is off.

Example:

```
<form action="" autocomplete="on ">
```

1. The <input> Element:

The HTML <input> element is the most used form element.

An <input> element can be displayed in many ways, depending on the type attribute.

Here are All types values:

- <input type="text">
- <input type="email">
- <input type="password">
- <input type="submit">
- <input type="reset">
- <input type="button">
- <input type="radio">
- <input type="checkbox">
- <input type="color">
- <input type="date">
- <input type="datetime-local">
- <input type="file">
- <input type="hidden">
- <input type="image">
- <input type="month">
- <input type="number">
- <input type="range">
- <input type="search">
- <input type="tel">
- <input type="time">
- <input type="url">
- <input type="week">

Input have more Attributes:

- **value** set a default value in input element
- **readonly** input will be readonly you cannot write some thing
- **required** input will be put something blank input can't submit
- **placeholder** show example in input tag
- **disabled** a set input disabled for not working
- **name** name specify the element name. in the backend it will use.
- **size** size specify the width on input.
- **maxlength** it is for limit the input field (text).
- **minlength** **i**t is for minimum limit
- **autofocus** input field should automatically get focus when the page loads
- **id** use with for attribute.
- **min** is limits the minimum value is.
- **max** is limits the maximum value is.
- **multiple** is only use in type=" file" for uploading two or more files.
- **step** if step="3", legal numbers could be -3, 0, 3, 6, etc.
- **pattern** regular expression that the input field's value is checked against.
- **Accept** it is accept only file extension.
- **list** refers to a <datalist> element that contains pre-defined options for an <input> element.

Note: Min & Max attributes work only with input types: number, range, date, datetime-local, month, time and week.

Example:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" size="50" placeholder=”Enter Name”
required ><br>
  <label for="lname">Last Name:</label><br>
<input type="text" id="lname" name="lname " maxlength="4" size="4" readonly>
</form>
```

Another Example:

```
<form>
  <label for="datemax">Enter a date before 1995-04-03:</label>
  <input type="date" id="datemax" name="datemax" max="1995-14-03"><br><br>

  <label for="datemin">Enter a date after 2000-01-01:</label>
  <input type="date" id="datemin" name="datemin" min="2000-01-02"><br><br>

 <label for="quantity">Quantity (between 1 and 5):</label>
 <input type="number" id="quantity" name="quantity" min="1" max="5">
</form>
```

Pattern Example:

```
<form>
  <label for="country_code">Country code:</label>
  <input type="text" id="country_code" name="country_code"
  pattern="[A-Za-z]{3}" title="Three letter country code">
</form>
```

List Attribute Example:

```
<form>
  <input list="browsers">

  <datalist id="browsers">
    <option value="Internet Explorer">
    <option value="Firefox">
    <option value="Chrome">
    <option value="Opera">
    <option value="Safari">
  </datalist>

</form>
```

Select and Option Element:

The <select> element is used to create a drop-down list.

The <option> tags inside the <select> element define the available options in the drop-down list.

Example:

```
<label for="country">Select Country:</label>

<select name="country" id="country">
  <option value="pakistan">Pakistan</option>
  <option value="india">India</option>
  <option value="bangladesh">Bangladesh </option>
  <option value="canada">Canada</option>
</select>
```

Select & Option with optgroup:

Select & option use with optgroup. This optgroup use for make option groups.

Example:

```
<label for="cars">Choose a car:</label>
<select name="cars" id="cars">
  <optgroup label="Japnese Cars">
    <option value="move">Move</option>
    <option value="mira">Mira</option>
    <option value="chr">CHR</option>
  </optgroup>
  <optgroup label="German Cars">
    <option value="mercedes">Mercedes</option>
    <option value="audi">Audi</option>
  </optgroup>
</select>
```

The textarea Element:

The <textarea> tag defines a multi-line text input control.
The <textarea> element is often used in a form, to collect user inputs like comments or reviews.

The size of a text area is specified by the cols and rows attributes (or with CSS).

Example:

```
<textarea id="comment" name="comment" rows="4" cols="20"></textarea>
```

The Button Element:

The <button> tag defines a clickable button.

Inside a <button> element you can put text (and tags like <i>, <b>, <strong>, <br>, <img>, etc.). That is not possible with a button created with the <input> element!

Example:

```
<button type="button">Click Me! </button>
```

The fieldset Element use with legend:

The <fieldset> tag is used to group related elements in a form.

The <fieldset> tag draws a box around the related elements.

Example:

```
<form>

<fieldset>
   <legend>Personal:</legend>
   <label for="fname">First name:</label>
   <input type="text" id="fname" name="fname"><br><br>
   <label for="lname">Last name:</label>
   <input type="text" id="lname" name="lname"><br><br>
   <input type="submit" value="Submit">
 </fieldset>

<fieldset>
   <legend>Academic Info:</legend>
   <label for="fname">First name:</label>
   <input type="text" id="fname" name="fname"><br><br>
   <label for="lname">Last name:</label>
   <input type="text" id="lname" name="lname"><br><br>
   <input type="submit" value="Submit">
 </fieldset></form>
```